FORMAL PATENT & COPYRIGHT SPECIFICATION

Intellectual Property & Technical Disclosure Document

Inventor, Applicant & Copyright Owner: Dhanshree Lonare
Jurisdiction: Patent Office (IPO / USPTO / PCT Framework)
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Section 1: Patent Application Specification

Title of Invention: System and Method for Real-Time Automated Malaria Parasite Detection and Clinical Triage Using Multi-Tier Convolutional Neural Networks

Field of the Invention: The present invention relates to medical diagnostic imaging systems, digital hematology, and computer-implemented deep convolutional neural network systems for automated detection, quantification, and clinical triage of *Plasmodium* malaria parasites in blood smear micrographs.

Background & Prior Art: Light microscopy of Giemsa-stained thin blood smears remains the clinical reference standard. However, manual screening is subjective, prone to high diagnostic discrepancy (25–40%), and constrained by human observer fatigue. Prior computerized systems relying on heuristic image processing (e.g. Otsu thresholding, watershed) fail under real-world staining variations and optical artifacts. Existing deep learning models lack edge-device adaptation and integrated clinical workflow deployment.

Summary of the Invention: The present invention provides a multi-tier clinical diagnostic system comprising an automated image ingestion engine, a multi-model deep learning inference pipeline (Custom CNN, edge-optimized MobileNetV2, and EfficientNetB0), and an interactive diagnostic web microservice delivering real-time confidence scores and triage tier assignments.

Patent Claims

I Claim:

1. A computer-implemented system for automated malaria parasite detection in blood smear micrographs, comprising a processor, an image ingestion module, and memory storing instructions to transform erythrocyte micrographs into normalized tensors, feed said tensors into a deep convolutional neural network, and generate real-time diagnostic classifications and confidence scores.

2. The system of claim 1, wherein said neural network comprises a four-stage hierarchical convolutional network with alternating Batch Normalization, Max-Pooling, and spatial Dropout regularisation ranging from 25% to 50%.

3. The system of claim 1, wherein said neural network comprises an edge-optimized MobileNetV2 architecture with depthwise separable convolutions configured to perform single-cell inference in under 20 milliseconds on a central processing unit.

4. The system of claim 1, further comprising an automated data augmentation module operative during training, applying randomized rotations between -20° and +20°, reflections, and zooming.

5. The system of claim 1, wherein said processor executes an inference web microservice communicating with an interactive browser-based glassmorphism diagnostic interface displaying real-time confidence ratings and clinical severity instructions.

6. The system of claim 1, achieving an Area Under the Receiver Operating Characteristic curve (AUC-ROC) exceeding 0.990 and sensitivity exceeding 96.5% on independent test sets.

7. A non-transitory computer-readable storage medium storing executable instructions to carry out the method of claim 1.

Section 2: Formal Copyright Declaration & Ownership

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The entire corpus of this project—including the algorithmic source code (app.py, malaria_detection.py, download_dataset.py), front-end user interface templates and stylesheets (templates/index.html), serialized deep learning neural network model weights (CustomCNN_best.h5, MobileNetV2_best.h5, EfficientNetB0_best.h5), benchmarking visualizations, and academic technical documentation—is the exclusive intellectual property of **Dhanshree Lonare**.

Declaration of Authorship:
I, Dhanshree Lonare, hereby certify that the architectural designs, empirical benchmarking methodologies, clinical user interface implementations, and technical specifications detailed herein are my original intellectual creations.

Signed and Executed:
Dhanshree Lonare
Lead Researcher, Developer & Patent Applicant
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